Functions and Tasks in Verilog HDL

Digital Design and RTL Fundamentals

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Contents

1 Introduction **2**

2 Functions in Verilog **2**
2.1 Basic Syntax . . . 2
2.2 Example 1: Add Two 4-bit Numbers . . . 2
2.3 Example 2: Even Parity Generator . . . 3
2.4 Example 3: Maximum of Two Values . . . 3
2.5 Example 4: 4-bit Population Count . . . 4

3 Important Rules for Functions **4**
3.1 Function with a Case Statement . . . 5

4 Tasks in Verilog **6**
4.1 Basic Syntax . . . 6
4.2 Example 1: Add and Subtract . . . 6
4.3 Example 2: Display a Formatted Message . . . 7
4.4 Example 3: Task with a Delay . . . 7
4.5 Example 4: Generate a Clock in a Testbench . . . 8

5 Functions and Tasks: Key Differences **8**

6 Static and Automatic Functions/Tasks **8**
6.1 Automatic Function Example . . . 8
6.2 Automatic Task Example . . . 9

7 Using Functions and Tasks in a Testbench **9**
7.1 Example: Self-Checking Combinational Adder . . . 10

8 Synthesis Considerations **11**

9 Common Errors **11**

10 Summary **12**

1 Introduction

Verilog HDL provides `function` and `task` constructs to group a sequence of statements under a name and reuse that behavior in a design or testbench. They help reduce repeated code and make descriptions easier to read, maintain, and verify.

A **function** is primarily used to calculate and return a value. A **task** is used to perform a procedure and can communicate through input, output, and inout arguments. In classic Verilog, both constructs are declared inside a module, and may be called from procedural code.

- Functions return one value and can be used inside expressions.
- Tasks can have multiple output values and are called as procedural statements.
- Both can accept input arguments.
- They are useful in RTL, behavioral models, and testbenches, subject to synthesis and timing restrictions.

2 Functions in Verilog

A function describes a calculation that produces a single result. In Verilog-2001 style, the function's name is also the variable to which the return value is assigned.

2.1 Basic Syntax

```
function [width-1:0] function_name;
    input [input_width-1:0] argument;
    begin
        function_name = expression;
    end
endfunction
```

The return width may be omitted for a one-bit result, but declaring it explicitly is often clearer. The function is called by writing its name with the required arguments, just like a value in an expression.

2.2 Example 1: Add Two 4-bit Numbers

```
module function_adder;
    reg [3:0] a, b;
    reg [4:0] sum;

    function [4:0] add4;
        input [3:0] x, y;
        begin
            add4 = x + y;
        end
    endfunction

```

```
    initial begin
        a = 4'd9;
        b = 4'd7;
        sum = add4(a, b);
        $display("a=%0d b=%0d sum=%0d", a, b, sum);
    end
endmodule
```

The function `add4` returns a 5-bit result so that the carry from adding two 4-bit unsigned values is not lost. The call `add4(a,b)` can be assigned to a variable or included in a larger expression.

2.3 Example 2: Even Parity Generator

```
module parity_example;
    reg [7:0] data;
    reg parity;

    function even_parity;
        input [7:0] x;
        begin
            even_parity = ^x;
        end
    endfunction

    initial begin
        data = 8'b10110010;
        parity = even_parity(data);
        $display("data=%b parity=%b", data, parity);
    end
endmodule
```

The reduction XOR operator ^ returns 1 when the input vector contains an odd number of ones. Appending this bit produces an even-parity word when the parity bit is included in the parity calculation.

2.4 Example 3: Maximum of Two Values

```
module maximum_example;
    reg [7:0] a, b, maximum;

    function [7:0] max8;
        input [7:0] x, y;
        begin
            if (x > y)
                max8 = x;
            else
                max8 = y;
        end
    endfunction
```

```
13
14      initial begin
15          a = 8'd45;
16          b = 8'd72;
17          maximum = max8(a, b);
18          $display("Maximum = %0d", maximum);
19      end
20  endmodule
```

This example compares unsigned 8-bit inputs. For signed comparisons, declare and interpret the operands consistently as signed values.

2.5 Example 4: 4-bit Population Count

A population-count function returns the number of logic-one bits in a vector.

```
1   module population_count_example;
2       reg [3:0] data;
3       reg [2:0] ones;
4
5       function [2:0] count_ones;
6           input [3:0] x;
7           integer i;
8           begin
9               count_ones = 0;
10              for (i = 0; i < 4; i = i + 1)
11                  count_ones = count_ones + x[i];
12          end
13      endfunction
14
15      initial begin
16          data = 4'b1101;
17          ones = count_ones(data);
18          $display("data=%b number_of_ones=%0d", data, ones);
19      end
20  endmodule
```

The loop has a fixed bound, which makes this coding style suitable for synthesis tools that support the construct. The function returns a 3-bit count because a 4-bit vector can contain zero through four ones.

3 Important Rules for Functions

Classic Verilog functions follow these important rules:

- A function returns one value, assigned to the function name.
- Function arguments are inputs; classic Verilog functions do not use output or inout arguments.

- A function must complete in zero simulation time. It cannot contain timing controls such as `#5`, event controls such as `@(posedge clk)`, or task calls that consume time.
- A function can be called from an expression, for example `y = f(a,b) + 1;`.
- Use a defined assignment to the function name on every intended execution path to avoid unintended unknown or stale results.
- Functions may contain procedural statements such as assignments, `if`, `case`, and bounded loops, provided the statements comply with the language and synthesis-tool restrictions.

3.1 Function with a Case Statement

```
module code_converter;
    reg [1:0] code;
    reg [3:0] output_value;

    function [3:0] decode_code;
        input [1:0] c;
        begin
            case (c)
                2'b00: decode_code = 4'b0001;
                2'b01: decode_code = 4'b0010;
                2'b10: decode_code = 4'b0100;
                2'b11: decode_code = 4'b1000;
                default: decode_code = 4'b0000;
            endcase
        end
    endfunction

    initial begin
        code = 2'b10;
        output_value = decode_code(code);
        $display("code=%b output=%b", code, output_value);
    end
endmodule
```

4 Tasks in Verilog

A task groups one or more procedural statements into a reusable procedure. Unlike a function, a task does not return a single expression value. It can pass information back using output or inout arguments, and it may contain timing controls when used in simulation.

4.1 Basic Syntax

```
task task_name;
    input  input_argument;
    output output_argument;
    reg    temporary_variable;
    begin
        // procedural statements
        output_argument = expression;
    end
endtask
```

A task is called as a statement. For example, use `task_name(a,b);` in a procedural block. A task may have only input arguments, or a combination of input, output, and inout arguments.

4.2 Example 1: Add and Subtract

```
module arithmetic_task;
    reg [7:0] a, b;
    reg [8:0] sum;
    reg [7:0] difference;

    task calculate;
        input  [7:0] x, y;
        output [8:0] s;
        output [7:0] d;
        begin
            s = x + y;
            d = x - y;
        end
    endtask

    initial begin
        a = 8'd25;
        b = 8'd10;
        calculate(a, b, sum, difference);
        $display("sum=%0d difference=%0d", sum, difference);
    end
endmodule
```

The task returns two results through its output arguments. The sum is 9 bits wide to retain the carry-out from unsigned 8-bit addition. The difference is 8 bits wide, so subtraction wraps modulo 256 for unsigned values.

4.3 Example 2: Display a Formatted Message

```
module display_task_example;
    reg [7:0] data;

    task show_data;
        input [7:0] value;
        begin
            $display("Decimal=%0d Hex=%h Binary=%b",
                     value, value, value);
        end
    endtask

    initial begin
        data = 8'hA5;
        show_data(data);
    end
endmodule
```

This task is intended for simulation and reporting. System tasks such as `$display` are not hardware logic and are generally not synthesizable.

4.4 Example 3: Task with a Delay

```
module delayed_task_example;
    reg signal;

    task drive_signal;
        input value;
        begin
            #5 signal = value;
        end
    endtask

    initial begin
        signal = 1'b0;
        drive_signal(1'b1);
        drive_signal(1'b0);
        #1 $finish;
    end
endmodule
```

The task waits five simulation time units before changing `signal`. The two calls therefore occur sequentially, with each call consuming simulation time. This is useful in testbenches but is not a synthesizable RTL pattern.

4.5 Example 4: Generate a Clock in a Testbench

```
module clock_task_example;
    reg clk;

    task wait_cycles;
        input integer count;
        integer i;
        begin
            for (i = 0; i < count; i = i + 1)
                #5 clk = ~clk;
        end
    endtask

    initial begin
        clk = 1'b0;
        wait_cycles(10);
        $display("Finished generating clock transitions");
        $finish;
    end
endmodule
```

This example toggles the clock every five time units for ten transitions. It is a testbench utility, not a clock-generation circuit to synthesize into hardware.

5 Functions and Tasks: Key Differences

Feature	Function	Task
Purpose	Calculate and return a value	Perform a procedure
Return mechanism	One value via function name	Output/inout arguments; no single expression return
Call style	Used in an expression	Called as a procedural statement
Timing controls	Not allowed in classic Verilog functions	Allowed by the language; common in testbenches
Simulation time	Must complete in zero time	May consume simulation time
Arguments	Input arguments	Input, output, and inout arguments
Typical use	Arithmetic, decode, parity, bit manipulation	Stimulus, checking, multi-result procedures

6 Static and Automatic Functions/Tasks

In classic Verilog, function and task variables are static by default: storage can persist between calls, and recursive or overlapping calls can therefore cause problems. The `automatic` keyword gives each invocation its own storage and is useful for re-entrant procedures and recursive routines where supported.

6.1 Automatic Function Example

```
module automatic_function_example;
    function automatic integer factorial;
        input integer n;
        begin
            if (n <= 1)
                factorial = 1;
            else
                factorial = n * factorial(n - 1);
        end
    endfunction

    initial begin
        $display("5! = %0d", factorial(5));
    end
endmodule
```

This recursive example uses an automatic function so that each recursive invocation has its own local argument and return storage. Tool support and recursion restrictions should be checked when synthesizing; this example is primarily for language learning and simulation.

6.2 Automatic Task Example

```
task automatic report_value;
    input integer index;
    input [7:0] value;
    begin
        $display("index=%0d value=%0d", index, value);
    end
endtask
```

This task can be declared within a module and called from a procedural block in that module. It is a simulation-reporting example.

7 Using Functions and Tasks in a Testbench

Functions and tasks can make testbenches more readable. A function can calculate an expected result, while a task can apply inputs, wait for the design to respond, compare values, and report the outcome.

7.1 Example: Self-Checking Combinational Adder

The following is a complete testbench with a combinational adder DUT and a checking task.

```
module adder_dut (
    input  [3:0] a, b,
    output [4:0] sum
);
    assign sum = a + b;
endmodule

module adder_testbench;
    reg [3:0] a, b;
    wire [4:0] sum;
    integer i, j;

    adder_dut dut (.a(a), .b(b), .sum(sum));

    task check_case;
        input [3:0] x, y;
        reg [4:0] expected;
        begin
            a = x;
            b = y;
            #1;
            expected = x + y;
            if (sum !== expected)
                $display("FAIL: a=%0d b=%0d expected=%0d actual=%0d",
                         x, y, expected, sum);
            else
                $display("PASS: a=%0d b=%0d sum=%0d", x, y, sum);
        end
    endtask

    initial begin
        for (i = 0; i < 16; i = i + 1)
            for (j = 0; j < 16; j = j + 1)
                check_case(i, j);
        $finish;
    end
endmodule
```

The task applies each test vector, waits one time unit for combinational settling, calculates the expected sum, and compares it with the DUT output. The case-inequality operator `!==` also treats unknown bits as a mismatch. In the actual project, compile both modules together in the simulator.

8 Synthesis Considerations

Functions and tasks are language constructs; whether a particular body is synthesizable depends on the statements used and the synthesis tool.

- Combinational functions using arithmetic, Boolean expressions, conditionals, case statements, and statically bounded loops are commonly synthesizable.
- A task used in synthesizable RTL may be accepted when it describes synthesizable logic and does not contain simulation-only behavior.
- Delays such as `#5`, event controls used as procedural waits, and system tasks such as `$display` are typically simulation-only and should not be used as hardware behavior in synthesizable RTL.
- Do not assume that wrapping non-synthesizable code inside a function or task makes it synthesizable.
- For sequential logic, use appropriate clocked processes and ensure that the function or task describes behavior compatible with the intended hardware.
- Check the supported Verilog version and synthesis-tool documentation, especially for automatic routines, recursion, and advanced language features.

9 Common Errors

1. **Forgetting to assign the function name:** every intended path should assign a value to the function name.
2. **Wrong return width:** size the function result to preserve carries or represent the full range of results.
3. **Calling a task as an expression:** tasks are invoked as procedural statements, not used as values in arithmetic expressions.
4. **Using delay controls in a function:** classic Verilog functions must complete in zero simulation time.
5. **Assuming all tasks are synthesizable:** delays and simulation system tasks inside a task remain non-synthesizable.
6. **Ignoring argument order:** task arguments are associated positionally in classic Verilog, so match the declared order and widths.
7. **Using blocking/nonblocking assignments carelessly:** use assignment types appropriate to the intended procedural behavior, especially in clocked RTL.

10 Summary

Functions and tasks provide a way to reuse procedural descriptions in Verilog HDL. Functions are best suited to calculations that return one value and can be used in expressions. Tasks are suited to procedures, including operations with multiple output arguments and simulation-time delays. Their use in synthesizable RTL depends on the constructs inside them, not simply on whether they are declared as a function or a task.